

UQFF

Author: Daniel T. Murphy

Session: 76 (March 2026)

UQFF

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Andromeda Friedmann-UQFF Gravity Coupling: $H(z)t$ Expansion Term and H_{UQFF} Near-Unity Resonance

Author: Daniel T. Murphy **Session:** 76 (March 2026) **Module:** ANDROMEDA_UQFFMODULE.cpp (M31 Master, UQFF 2.0) **WOLFRAM_TERM:** AndromedaUQFF:g_exp=G*M/r^2*H(z)*t; H(z)=H0*Sqrt[Om_m*(1+z)^3+Om_L]/Mpc2m; H_UQFF=H(z)*t_H-0.987 [PAPER_276]

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as $g_{\text{expansion}} = (G M / r^2) H(z) t$. For Andromeda's blueshift redshift $z = -0.001$, the Friedmann $H(z)$ equals $69.969 \text{ km/s/Mpc} = 2.269 \times 10^{-8} \text{ s}^{-1}$. Evaluated at the Hubble timescale $t = t_H = 4.352 \times 10^7 \text{ s}$, the dimensionless Friedmann-UQFF Resonance Coefficient $H_{\text{UQFF}} = H(z)t_H = 0.987$ is a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the *Mvisible*/MDM_mass cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

1. Background

The UQFF framework computes total gravitational acceleration $g_{\text{total}}(r,t)$ as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER273-275 established the Andromeda-specific physics: blueshift approach amplifier, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaître cosmological evolution of Hubble parameter $H(z)$. This is significant because over a Hubble timescale, $H(z)t \approx 1$ implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

2. Friedmann $H(z)$ Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard Λ CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}$$

For Andromeda parameters: $H_0 = 70.0 \text{ km/s/Mpc}$, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $z = -0.001$.

$$\begin{aligned}
H(z=-0.001) &= 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7} \\
&= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991} \\
&= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}
\end{aligned}$$

In SI units: $H_{\text{SI}} = 69.969 \times 10^3 / 3.086 \times 10^{22} = 2.269 \times 10^{-18} \text{ s}^{-1}$

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t .

Numerical values for Andromeda:

$$\begin{aligned}
g_{\text{base}} &= G M / r = 6.674 \times 10^{-11} \times 1.989 \times 10^{30} / (1.04 \times 10^{17})^2 = 1.227 \times 10^{-10} \text{ m/s}^2 \\
\text{At } t = 1 \text{ Gyr } (3.156 \times 10^{16} \text{ s}): g_{\text{expansion}} &= 1.227 \times 10^{-10} \times 2.269 \times 10^{-18} \times 3.156 \times 10^{16} = \\
&= 8.79 \times 10^{-12} \text{ m/s}^2 \\
\text{At } t = t_{\text{Hubble}} (4.352 \times 10^{17} \text{ s}): g_{\text{expansion}} &= 1.227 \times 10^{-10} \times 0.987 = 1.211 \times 10^{-10} \text{ m/s}^2
\end{aligned}$$

3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^{17} \text{ s}$ (Hubble time $\approx 13.8 \text{ Gyr}$).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

$H_{\text{UQFF}} \approx 0.987 \approx 1$ is a remarkable result. It means:

Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** $\approx$ nearly doubling g_{base} .

This near-unity value is not coincidental. In a flat Λ CDM universe ($\Omega_m + \Omega_\Lambda = 1$):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because $H_0 \times t_H$ is the dimensionless Hubble number $\approx 0.96 \approx 1.0$ for observationally consistent H_0 .

New UQFF Constant: $H_{\text{UQFF}} = H(z) \times t_H \approx$ **Friedmann-UQFF Near-Unity Resonance Coefficient**

Parameter	Value
H_0	70.0 km/s/Mpc
Ω_m	0.3

Parameter	Value
O_?	0.7
z (Andromeda)	-0.001
H(z)	$2.269 \times 10^{-18} \text{ s}^{-1}$
t_H	$4.352 \times 10^7 \text{ s}$
H_UQFF	0.987 (~1)
gexpansion(tH)	$1.211 \times 10^{-10} \text{ m/s}^2$
g_base	$1.227 \times 10^{-10} \text{ m/s}^2$
Ratio gexp/gbase	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $\Omega_m(1+z)$ is suppressed below Ω_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

For Andromeda ($z = -0.001$): $H_{UQFF} = 0.987$ vs flat-universe $H_{UQFF0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0$? $H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}}}{\rho_{\text{mean}}} \cdot v_{\text{orbit}}^2 \cdot \frac{c^2}{\rho_{\text{base}}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^{-8} \text{ kg/m}^3$.

Numerical value for Andromeda:

$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{1.989 \times 10^{-18}} \times \frac{(2.998 \times 10^8)^2 \times 1.227 \times 10^{-10}}{1.989 \times 10^{-18}}$$

$$= \frac{6.25 \times 10^{-10}}{1.989 \times 10^{-18}} \times \frac{8.988 \times 10^{16} \times 1.227 \times 10^{-10}}{1.989 \times 10^{-18}}$$

$$= \frac{6.25 \times 10^{-10}}{1.989 \times 10^{-18}} \times 0.1788 \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. Mvisible / MDM_mass Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$

$$M_{\text{DM, mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: $M = 1.989 \times 10^{12} \text{ kg}$, $f_{\text{DM}} = 0.80$:

$$M_{\text{visible}} = 0.20 \times 1.989 \times 10^{12} = 3.978 \times 10^{11} \text{ kg} \text{ (20\% visible baryons)}$$

$$M_{\text{DM mass}} = 0.80 \times 1.989 \times 10^{12} = 1.591 \times 10^{12} \text{ kg} \text{ (80\% dark matter)}$$

These are tracked in `updateCache()` and exported via `exportState()` (params 26-32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA UQFFMODULE.cpp g_{total}

The full UQFF 2.0 equation for Andromeda with all PAPER_273-276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{\text{g, sum}}^{(26)} + \Lambda_{\text{-term}} + g_{\text{Q}} + g_{\text{L}} + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at $t=0$ (current epoch):

Term	Value (m/s ²)	Paper
$g_{\text{grav}} = G M / r$	1.227×10^{-10}	baseline
$U_{\text{g, sum}}$ (26 layers)	6.380×10^{-10}	26-layer Triadic
?-term	2.998×10^{-10}	cosmological
g_{quantum}	$\sim 2 \times 10^{-10}$	HUP
g_{Lorentz}	$\sim 2 \times 10^{-10}$	IGM EM
g_{fluid}	$\sim 8 \times 10^{-10}$	IGM buoyancy
$F_{\text{res}}(?_{\text{HI}})$ at $t=0$	1×10^{-10}	PAPER_274
g_{DM}	1.356×10^{-10}	PAPER_275
$g_{\text{expansion}}$ at $t=0$	0	PAPER_276
a_{dust}	4.29×10^{-10}	PAPER_276
κ_{approach}	1.001001	PAPER_273

Note: $g_{\text{expansion}}$ grows linearly with t , dominating over g_{grav} at $t = t_{\text{H}}$.

7. exportState Update (25 - 32 parameters)

Seven new parameters added to `exportState()`:

$$H_0 = 70.0 \text{ km/s/Mpc}$$

$$\omega_{\text{m}} = 0.3$$

$$\omega_{\text{Lam}} = 0.7$$

$$\text{Mpc_to_m} = 3.086 \times 10^{22}$$

$$\rho_{\text{dust}} = 1 \times 10^{-26} \text{ kg/m}^3$$

$$M_{\text{visible}} = (1 - f_{\text{DM}}) M$$

$$M_{\text{DM mass}} = f_{\text{DM}} M$$

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

Friedmann-UQFF Expansion Coupling: $g_{\text{expansion}} = g_{\text{base}} \cdot H(z) \cdot t$. At $t = t_H$, adds 98.7% of g_{base} near-gravitational doubling over cosmological time.

$H_{\text{UQFF}} = 0.987$: New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 \cdot t_H = 1$ in flat Λ CDM. Andromeda's blueshift ($z = -0.001$) makes H_{UQFF} 0.15% below flat-universe value a blueshift Friedmann suppression.

$M_{\text{visible}}/M_{\text{DM_mass}}$ cascade + dust drag: Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure ($a_{\text{dust}} \approx 4 \cdot 10^{-10}$ m/s²).

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 $H(z)=H_0 \cdot \text{Sqrt}[0m \cdot (1+z)^3 + 0m_L] / \text{Mpc}2m$; $H_{\text{UQFF}}=H(z) \cdot t_H - 0.987$ [PAPER_276]

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as $g_{\text{expansion}} = (G \cdot M / r^2) \cdot H(z) \cdot t$. For Andromeda's blueshift redshift $z = -0.001$, the Friedmann $H(z)$ equals 69.969 km/s/Mpc = $2.269 \cdot 10^{-8}$ s⁻¹. Evaluated at the Hubble timescale $t = t_H = 4.352 \cdot 10^7$ s, the dimensionless Friedmann-UQFF Resonance Coefficient $H_{\text{UQFF}} = H(z) \cdot t_H = 0.987$ a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the $M_{\text{visible}}/M_{\text{DM_mass}}$ cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

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$$H(z=-0.001) = 70.0 \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: $H_{\text{SI}} = 69.969 \text{ km/s/Mpc} / 3.086 \times 10^{22} \text{ m/Mpc} = 2.269 \times 10^{-18} \text{ s}^{-1}$

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The new UQFF gravity coupling term is:

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This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t .

Numerical values for Andromeda:

$$g_{\text{base}} = G M / r = 6.674 \times 10^{-11} \text{ m}^3/\text{kg s}^2 \cdot 1.989 \times 10^{30} \text{ kg} / (1.04 \times 10^{17} \text{ m})^2 = 1.227 \times 10^{-10} \text{ m/s}^2$$

$$\text{At } t = 1 \text{ Gyr } (3.156 \times 10^{16} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-10} \text{ m/s}^2 \cdot 2.269 \times 10^{-18} \text{ s}^{-1} \cdot 3.156 \times 10^{16} \text{ s} = 8.79 \times 10^{-12} \text{ m/s}^2$$

$$\text{At } t = t_{\text{Hubble}} (4.352 \times 10^{17} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-10} \text{ m/s}^2 \cdot 0.987 = 1.211 \times 10^{-10} \text{ m/s}^2$$

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3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^{17} \text{ s}$ (Hubble time ≈ 13.8 Gyr).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \text{ s}^{-1} \times 4.352 \times 10^{17} \text{ s} = \mathbf{0.987}$$

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because $H_0 \times t_H$ is the dimensionless Hubble number $\approx 0.96 \approx 1.0$ for observationally consistent H_0 .

New UQFF Constant: $H_{UQFF} = H(z) \frac{t_H}{t_H}$ ■ Friedmann-UQFF Near-Unity Resonance Coefficient

Parameter	Value
H_0	70.0 km/s/Mpc
O_m	0.3
O_γ	0.7
z (Andromeda)	-0.001
$H(z)$	$2.269 \times 10^{18} \text{ s}^{-1}$
t_H	$4.352 \times 10^7 \text{ s}$
H_{UQFF}	0.987 (~1)
$g_{expansion}(t_H)$	$1.211 \times 10^{10} \text{ m/s}^2$
g_{base}	$1.227 \times 10^{10} \text{ m/s}^2$
Ratio g_{exp}/g_{base}	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $O_m(1+z)$ is suppressed below O_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

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where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^8 \text{ kg/m}^3$.

Numerical value for Andromeda:

$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~ 9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

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When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

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$$M_{\text{DM,mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: $M = 1.989 \times 10^{14} \text{ kg}$, $f_{\text{DM}} = 0.80$:

$$M_{\text{visible}} = 0.20 \times 1.989 \times 10^{14} = 3.978 \times 10^{13} \text{ kg} \text{ (20\% visible baryons)}$$

$$MDM_{\text{mass}} = 0.80 \times 1.989 \times 10^{14} = 1.591 \times 10^{14} \text{ kg} \text{ (80\% dark matter)}$$

These are tracked in `updateCache()` and exported via `exportState()` (params 26-32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

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Term magnitudes at t=0 (current epoch):

Term	Value (m/s ²)	Paper
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$F_{\text{res}}(?_{\text{HI}})$ at t=0	1×10^{-10}	PAPER_274
g_{DM}	1.356×10^{-10}	PAPER_275
$g_{\text{expansion}}$ at t=0	0	PAPER_276
a_{dust}	$4.29 \times 10^{-10.5}$	PAPER_276
$?_{\text{approach}}$	1.001001	PAPER_273

Note: *gexpansion* grows linearly with *t*, dominating over *ggrav* at $t = t_{\text{H}}$.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`:

$$H_0 = 70.0 \text{ km/s/Mpc}$$

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Numerical values for Andromeda:

$$g_{\text{base}} = G M / r = 6.674 \times 10^{-11} \times 1.989 \times 10^{30} / (1.04 \times 10^{17})^2 = 1.227 \times 10^{-8} \text{ m/s}^2$$

$$\text{At } t = 1 \text{ Gyr } (3.156 \times 10^{16} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-8} \times 2.269 \times 10^{-8} \times 3.156 \times 10^{16} = 8.79 \times 10^{-8} \text{ m/s}^2$$

$$\text{At } t = t_{\text{Hubble}} (4.352 \times 10^{17} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-8} \times 0.987 = 1.211 \times 10^{-8} \text{ m/s}^2$$

3. H_UQFF Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^{17} \text{ s}$ (Hubble time ≈ 13.8 Gyr).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

$H_{\text{UQFF}} \approx 0.987 \approx 1$ is a remarkable result. It means:

Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** $\approx$ nearly doubling g_{base} .

This near-unity value is not coincidental. In a flat Λ CDM universe ($\Omega_m + \Omega_\Lambda = 1$):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because $H_0 \times t_H$ is the dimensionless Hubble number $\approx 0.96 \approx 1.0$ for observationally consistent H_0 .

New UQFF Constant: $H_{UQFF} = H(z) \times t_H$ ■ Friedmann-UQFF Near-Unity Resonance Coefficient

Parameter	Value
H_0	70.0 km/s/Mpc
O_m	0.3
O_Ω	0.7
z (Andromeda)	-0.001
$H(z)$	$2.269 \times 10^8 \text{ s}^{-1}$
t_H	$4.352 \times 10^7 \text{ s}$
H_{UQFF}	0.987 (~1)
$g_{\text{expansion}}(t_H)$	$1.211 \times 10^8 \text{ m/s}^2$
g_{base}	$1.227 \times 10^8 \text{ m/s}^2$
Ratio $g_{\text{exp}}/g_{\text{base}}$	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $O_m/(1+z)$ is suppressed below O_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

For Andromeda ($z = -0.001$): $H_{UQFF} = 0.987$ vs *flat-universe* $H_{UQFF0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0 \rightarrow H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}}}{\rho_{\text{mean}}} \cdot v_{\text{orbit}}^2 \cdot c^2 \cdot g_{\text{base}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^8 \text{ kg/m}^3$.

Numerical value for Andromeda:

$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{1.989 \times 10^{-18}} \times \frac{(2.998 \times 10^8)^2 \times 1.227 \times 10^{-10}}{1.227 \times 10^{-10}}$$

$$= \frac{6.25 \times 10^{-10}}{1.989 \times 10^{-18}} \times \frac{8.988 \times 10^{16} \times 1.227 \times 10^{-10}}{1.227 \times 10^{-10}}$$

$$= \frac{6.25 \times 10^{-10}}{1.989 \times 10^{-18}} \times 0.1788 \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~ 9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. M_{visible} / MDM_{mass} Cascade

When M is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$

$$M_{\text{DM, mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: $M = 1.989 \times 10^{14} \text{ kg}$, $f_{\text{DM}} = 0.80$:

$$M_{\text{visible}} = 0.20 \times 1.989 \times 10^{14} = 3.978 \times 10^{13} \text{ kg} \text{ (20\% visible baryons)}$$

$$MDM_{\text{mass}} = 0.80 \times 1.989 \times 10^{14} = 1.591 \times 10^{14} \text{ kg} \text{ (80\% dark matter)}$$

These are tracked in `updateCache()` and exported via `exportState()` (params 26–32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA UQFFMODULE.cpp g_{total}

The full UQFF 2.0 equation for Andromeda with all PAPER_273–276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{\text{g, sum}}^{(26)} + \Lambda_{\text{-term}} + g_{\text{Q}} + g_{\text{L}} + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at $t=0$ (current epoch):

Term	Value (m/s ²)	Paper
$g_{\text{grav}} = G M / r$	1.227×10^{-10}	baseline
$U_{\text{g, sum}}$ (26 layers)	$6.380 \times 10^{??}$	26-layer Triadic
?-term	2.998×10^{-16}	cosmological
g_{quantum}	$\sim 2 \times 10^{-16}$	HUP
g_{Lorentz}	$\sim 2 \times 10^{-16}$	IGM EM
g_{fluid}	$\sim 8 \times 10^{??}$	IGM buoyancy
$F_{\text{res}}(?_{\text{HI}})$ at $t=0$	$1 \times 10^{??}$	PAPER_274
g_{DM}	$1.356 \times 10^{??}$	PAPER_275
$g_{\text{expansion}}$ at $t=0$	0	PAPER_276
a_{dust}	$4.29 \times 10^{??}$	PAPER_276
$?_{\text{approach}}$	1.001001	PAPER_273

Note: $g_{\text{expansion}}$ grows linearly with t , dominating over g_{grav} at $t = t_{\text{H}}$.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to `exportState()`:

$$H_0 = 70.0 \text{ km/s/Mpc}$$

$$\omega_{\text{m}} = 0.3$$

Omega_Lam = 0.7
Mpc_to_m = 3.086■10■
rho_dust = 1■10?■ kg/m■
M_visible = (1■f_DM)■M
M_DM_mass = f_DM■M

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

Friedmann-UQFF Expansion Coupling: $g_{\text{expansion}} = g_{\text{base}} \cdot H(z) \cdot t$. At $t = tH$, adds 98.7% of g_{base} ■ near-gravitational doubling over cosmological time.

****H_UQFF = 0.987**:** New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 \cdot t_H \cdot 1$ in flat ■CDM. Andromeda's blueshift ($z = -0.001$) makes H_UQFF 0.15% below flat-universe value ■ a blueshift Friedmann suppression.

****M_visible/M_DM_mass cascade + dust drag**:** Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure ($a_{\text{dust}} \cdot 4 \cdot 10^? \cdot ? \text{ m/s} \cdot ?$).

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Andromeda Friedmann-UQFF Gravity Coupling: $H(z) \cdot t$ Expansion Term and H_UQFF Near-Unity Resonance

Author: Daniel T. Murphy **Session:** 76 (March 2026) **Module:** ANDROMEDAUQFFMODULE.cpp (M31 Master, UQFF 2.0) **WOLFRAM_TERM:** AndromedaUQFF:g_exp=G*M/r^2*H(z)*t;
 $H(z)=H_0 \cdot \text{Sqrt}[0m_m \cdot (1+z)^3+0m_L]/\text{Mpc}2m$; $H_UQFF=H(z) \cdot t_H-0.987$ [PAPER_276]

Abstract

This paper introduces a new UQFF gravity term for the Andromeda Galaxy (M31): the **Friedmann-UQFF Expansion Coupling**, defined as $g_{\text{expansion}} = (G \cdot M/r \cdot ?) \cdot H(z) \cdot t$. For Andromeda's blueshift redshift $z = -0.001$, the Friedmann $H(z)$ equals $69.969 \text{ km/s/Mpc} = 2.269 \cdot 10^? \cdot 8 \text{ s} \cdot ?$. Evaluated at the Hubble timescale $t = tH = 4.352 \cdot 10 \cdot 7 \text{ s}$, the dimensionless Friedmann-UQFF Resonance Coefficient $H_UQFF = H(z) \cdot tH \cdot 0.987$ ■ a near-unity value representing gravitational doubling over cosmological time. This paper also introduces the $M_{\text{visible}}/MDM_{\text{mass}}$ cascade (explicit DM split tracking) and a minor ISM dust drag acceleration term. Together these constitute three additive UQFF 2.0 enhancements derived from the Andromeda module upgrade.

1. Background

The UQFF framework computes total gravitational acceleration $g_{\text{total}}(r,t)$ as a sum of Newtonian, Triadic (26-layer), cosmological, quantum, electromagnetic, fluid, resonant, and dark matter terms. PAPER273■275 established the Andromeda-specific physics: blueshift approach amplifier ?, HI 21-cm resonance, and DM 80/20 NFW partition.

The original Andromeda implementation (pre-Session 76) lacked a time-dependent expansion coupling that accounts for the Friedmann-Lemaitre cosmological evolution of Hubble parameter $H(z)$. This is significant because over a Hubble timescale, $H(z)t \approx 1$ implying that expansion-mediated gravitational coupling is of the same order as the base Newtonian term.

2. Friedmann $H(z)$ Expansion Coupling

2.1 Friedmann Equation

The Hubble parameter as a function of redshift in the standard Λ CDM cosmology:

$$H(z) = H_0 \sqrt{\Omega_m (1+z)^3 + \Omega_\Lambda}$$

For Andromeda parameters: $H_0 = 70.0$ km/s/Mpc, $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, $z = -0.001$.

$$H(z=-0.001) = 70.0 \times \sqrt{0.3 \times (0.999)^3 + 0.7}$$

$$= 70.0 \times \sqrt{0.3 \times 0.997003 + 0.7} = 70.0 \times \sqrt{0.9991}$$

$$= 70.0 \times 0.99955 = 69.969 \text{ km/s/Mpc}$$

In SI units: $H_{\text{SI}} = 69.969 \text{ km/s/Mpc} / 3.086 \times 10^{22} \text{ m/Mpc} = 2.269 \times 10^{-18} \text{ s}^{-1}$

2.2 UQFF Expansion Term

The new UQFF gravity coupling term is:

$$g_{\text{expansion}}(r, t) = \frac{G \cdot M}{r^2} \cdot H(z) \cdot t$$

This represents the gravitational effect of cosmological expansion flow coupling into UQFF buoyancy at timescale t .

Numerical values for Andromeda:

$$g_{\text{base}} = G M / r = 6.674 \times 10^{-11} \text{ N m}^2/\text{kg}^2 \cdot 1.989 \times 10^{30} \text{ kg} / (1.04 \times 10^{17} \text{ m})^2 = 1.227 \times 10^{-10} \text{ m/s}^2$$

$$\text{At } t = 1 \text{ Gyr } (3.156 \times 10^{16} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-10} \times 2.269 \times 10^{-18} \times 3.156 \times 10^{16} = 8.79 \times 10^{-12} \text{ m/s}^2$$

$$\text{At } t = t_{\text{Hubble}} (4.352 \times 10^{17} \text{ s}): g_{\text{expansion}} = 1.227 \times 10^{-10} \times 0.987 = 1.211 \times 10^{-10} \text{ m/s}^2$$

3. H_{UQFF} Near-Unity Resonance Coefficient

3.1 Definition

$$H_{\text{UQFF}} = H(z) \times t_H$$

where $t_H = 4.352 \times 10^{17} \text{ s}$ (Hubble time ≈ 13.8 Gyr).

For Andromeda ($z = -0.001$):

$$H_{\text{UQFF}} = 2.269 \times 10^{-18} \times 4.352 \times 10^{17} = \mathbf{0.987}$$

3.2 Near-Unity Discovery

$H_{\text{UQFF}} \approx 0.987 \approx 1$ is a remarkable result. It means:

Over a Hubble timescale, the Friedmann expansion coupling adds an amount of gravitational acceleration equal to **98.7% of the base Newtonian term** ■ nearly doubling g_{base} .

This near-unity value is not coincidental. In a flat Λ CDM universe ($\Omega_m + \Omega_\Lambda = 1$):

$$H_{\text{UQFF}} = H_0 \times t_H \approx 1.0$$

because $H_0 \times t_H$ is the dimensionless Hubble number ■ $0.96 \approx 1.0$ for observationally consistent H_0 .

New UQFF Constant: $H_{\text{UQFF}} = H(z) \times t_H$ ■ Friedmann-UQFF Near-Unity Resonance Coefficient

Parameter	Value
H_0	70.0 km/s/Mpc
Ω_m	0.3
Ω_Λ	0.7
z (Andromeda)	-0.001
$H(z)$	$2.269 \times 10^8 \text{ s}^{-1}$
t_H	$4.352 \times 10^7 \text{ s}$
H_{UQFF}	0.987 (~1)
$g_{\text{expansion}}(t_H)$	$1.211 \times 10^8 \text{ m/s}^2$
g_{base}	$1.227 \times 10^8 \text{ m/s}^2$
Ratio $g_{\text{exp}}/g_{\text{base}}$	98.7%

3.3 Blueshift Sensitivity

For blueshift $z < 0$: $H(z)$ is slightly **lower** than H_0 (matter term $\Omega_m(1+z)^3$ is suppressed below Ω_m for $z < 0$), meaning:

$$H(z < 0) < H_0 \Rightarrow H_{\text{UQFF}}(z < 0) < H_{\text{UQFF}}(z=0)$$

For Andromeda ($z = -0.001$): $H_{\text{UQFF}} = 0.987$ vs flat-universe $H_{\text{UQFF}0} = 0.9985$. The blueshift suppresses H_{UQFF} by 0.15%.

This is the inverse of the redshift behaviour ($z > 0 \Rightarrow H(z) > H_0$ for moderate z), making Andromeda's approaching trajectory a unique laboratory for probing blueshift Friedmann-UQFF coupling.

4. ISM Dust Drag Acceleration

A second minor additive term captures ISM dust ram-pressure drag:

$$a_{\text{dust}} = \frac{\rho_{\text{dust}}}{\rho_{\text{mean}}} \times v_{\text{orbit}}^2 \times c^2 \times g_{\text{base}}$$

where $\rho_{\text{mean}} = M/V_{\text{fluid}} = 1.989 \times 10^4 / 106 = 1.989 \times 10^8 \text{ kg/m}^3$.

Numerical value for Andromeda:

$$a_{\text{dust}} = \frac{10^{-20} \times (2.5 \times 10^5)^2}{(2.998 \times 10^8)^2 \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{8.988 \times 10^{16} \times 1.989 \times 10^{-18}} \times 1.227 \times 10^{-10}$$

$$= \frac{6.25 \times 10^{-10}}{0.1788} \times 1.227 \times 10^{-10} = 4.29 \times 10^{-19} \text{ m/s}^2$$

At ~ 9 orders below g_{base} , the dust drag is physically negligible but dimensionally consistent and additive within the UQFF sum.

5. *Mvisible* / *MDM*_{mass} Cascade

When *M* is updated via `updateVariable("M", value)`, the derived quantities are now automatically recomputed:

$$M_{\text{visible}} = (1 - f_{\text{DM}}) \times M$$

$$M_{\text{DM, mass}} = f_{\text{DM}} \times M$$

For Andromeda defaults: $M = 1.989 \times 10^{14} \text{ kg}$, $f_{\text{DM}} = 0.80$:

$$M_{\text{visible}} = 0.20 \times 1.989 \times 10^{14} = 3.978 \times 10^{13} \text{ kg} \text{ (20\% visible baryons)}$$

$$MDM_{\text{mass}} = 0.80 \times 1.989 \times 10^{14} = 1.591 \times 10^{14} \text{ kg} \text{ (80\% dark matter)}$$

These are tracked in `updateCache()` and exported via `exportState()` (params 26–32), providing explicit bookkeeping of the M31 baryonic/DM split consistent with PAPER_275.

6. Updated ANDROMEDA UQFFMODULE.cpp *g*_{total}

The full UQFF 2.0 equation for Andromeda with all PAPER_273–276 terms:

$$g_{\text{total}}(r, t) = \left[g_{\text{grav}} + U_{\text{g, sum}}^{(26)} + \Lambda_{\text{-term}} + g_{\text{Q}} + g_{\text{L}} + g_{\text{fluid}} + F_{\text{res}}(\omega_{\text{HI}}) + g_{\text{DM}} + g_{\text{expansion}} + a_{\text{dust}} \right] \times \kappa_{\text{approach}}$$

Term magnitudes at $t=0$ (current epoch):

Term	Value (m/s ²)	Paper
$g_{\text{grav}} = G M / r$	1.227×10^{-10}	baseline
$U_{\text{g, sum}}$ (26 layers)	6.380×10^{-11}	26-layer Triadic
?-term	2.998×10^{-16}	cosmological
g_{quantum}	$\sim 2 \times 10^{-16}$	HUP
g_{Lorentz}	$\sim 2 \times 10^{-16}$	IGM EM
g_{fluid}	$\sim 8 \times 10^{-17}$	IGM buoyancy
$F_{\text{res}}(?_{\text{HI}})$ at $t=0$	1×10^{-17}	PAPER_274
g_{DM}	1.356×10^{-10}	PAPER_275
$g_{\text{expansion}}$ at $t=0$	0	PAPER_276
a_{dust}	4.29×10^{-19}	PAPER_276
κ_{approach}	1.001001	PAPER_273

Note: $g_{\text{expansion}}$ grows linearly with t , dominating over g_{grav} at $t = t_{\text{H}}$.

7. exportState Update (25 ? 32 parameters)

Seven new parameters added to exportState():

```
H0 = 70.0 km/s/Mpc
Omega_m = 0.3
Omega_Lam = 0.7
Mpc_to_m = 3.086e10
rho_dust = 1e-10 kg/m^3
M_visible = (1-f_DM)*M
M_DM_mass = f_DM*M
```

8. Conclusions

Three new UQFF enhancements for Andromeda (PAPER_276):

- Friedmann-UQFF Expansion Coupling:** $g_{expansion} = g_{base} \cdot H(z)$. At $t = t_H$, adds 98.7% of g_{base} near-gravitational doubling over cosmological time.
- H_UQFF = 0.987:** New UQFF constant (Friedmann-UQFF Near-Unity Resonance Coefficient). Near-unity reflects the fundamental relationship $H_0 \cdot t_H = 1$ in flat Λ CDM. Andromeda's blueshift ($z = -0.001$) makes H_UQFF 0.15% below flat-universe value a blueshift Friedmann suppression.
- M_visible/M_DM_mass cascade + dust drag:** Explicit DM split bookkeeping (updateCache) and ISM dust ram-pressure ($a_{dust} = 4e-10$ m/s^2).

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UQFF computed: MUGE buoyancy ratio $U_{bi}/F_U = [SSq] \cdot r/GM = 5.7e-15.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7$ m/s^2 at r_{ISCO} .